

Programme	BSc. Computer Science				
Course Title	Statistical Foundations for Artificial Intelligence				
Type of Course	Vocational Minor				
Semester	I				
Academic Level	100 - 199				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	1. A strong foundation in algebra 2. Fundamentals of Set theory and logic				
Course Summary	The course on probability and statistics covers fundamental topics including descriptive statistics (measures of central tendency and dispersion), probability theory (events, sample spaces, probability laws, random variables, and distributions), inferential statistics (regression analysis), and applications in various fields such as science, engineering, economics, and social sciences, emphasizing critical thinking, data analysis, and problem-solving skills.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Apply fundamental statistics concepts	Ap	C	Quizzes, Homework, Exams
CO2	Analyze data using descriptive statistics	An	P	Projects, Midterm, Exams
CO3	Perform regression analysis	An	P	Projects, Exams
CO4	Apply probability and statistics in real-world situations	Ap	C	Projects, Exams
CO5	Develop critical thinking and problem-solving skills	E	M	Homework, Projects
CO6	Communicate statistical findings effectively	E	M	Presentations, Reports
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus

Module	Unit	Content	Hrs (45+30)	Mark
I	DESCRIPTIVE STATISTICS		9	15
	1	Concept of primary and secondary data, Methods of collection	1	
	2	Measures of central tendencies (Mean, Median, Mode, HM, GM)	4	
	3	Measures of dispersion, Relative Measures and Absolute Measures	2	
	4	Range, Quartile deviation, Mean deviation, standard deviation, Variance	2	
II	STATISTICAL INFERENCE AND REGRESSION ANALYSIS		11	15
	5	Principles of Least Squares and Fitting of Stright Line	2	
	6	Point estimation: maximum likelihood estimation (MLE), method of moments. Confidence intervals for population parameters.	3	
	7	Pearson's Coefficient of Correlation and Rank Correlation	3	
	8	Simple linear regression and multiple linear regression. Logistic regression for classification problems.	3	
III	PROBABILITY THEORY		11	20
	9	Random experiment, Sample point, Sample Space	1	
	10	Events, Operation of events (Union, Intersection, Complement of Events)	2	
	11	Exclusive and exhaustive events, equally likely events with examples	1	
	12	Classical approach to probability	1	
	13	Axiomatic definitions of probability, simple problems	2	
	14	Conditional probability	1	
	15	Inverse probability	1	
	16	Baye's Theorem	2	
IV	ADVANCED PROBABILITY DISTRIBUTION		14	20
	17	Discrete and continuous random variables and probability distribution	2	
	18	Binomial distribution: Definition, Expectation, Variance, Moment Generating Function and Problems	2	
	19	Poisson distribution: Definition, Expectation, Variance, Moment Generating Function and Problems	2	
	20	Normal distribution: Definition, Expectation, Variance, Moment Generating Function, Standard normal curve and Problems	3	

	21	Testing of Hypothesis: General principles of testing, Two types of errors	3	
	22	Type of Testing: T-Test, ANOVA-Test, Chi-square test (Concept Only)	2	
V	Lab Activities (Use Sci Lab)		30	
	1	Implements mean, median and mode hight of then students	20	
		Determine the standard deviation and variance		
		Plot a histogram to visualize their distribution		
		Use SciLab to perform simple linear regression on a dataset with two variables.		
		Implement SciLab code to plot box plots, scatter plots, and density plots for the dataset to explore its characteristics.		
		Use SciLab to perform Least Square		
		Implement algorithms for multiple linear regression and logistic regression in SciLab to predict outcomes based on input features.		
		You have a deck of 52 playing cards. Calculate the probability of drawing a face card (jack, queen, or king) from the deck.		
		Simulate random experiments and calculate probabilities of events using Scilab.		
		Write functions in SciLab to calculate probabilities for events based on given probability distributions (e.g., binomial, normal).		
	2	Case Study	2	
	3	Develop a predictive model using statistical techniques and tools for identifying a real-world problem in Artificial Intelligence.	8	

Mapping of COs with PSOs and POs:

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
CO 1	1	3	-	1	1	1						
CO 2	1	3	-	-	1	-						
CO 3	1	3	-	-	2	2						
CO 4	1	3	-	-	2	2						
CO 5	2	1	-	1	1	-						
CO 6	2	1	1	2	2	1						

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Practical Evaluation	End Semester Examinations
CO 1	✓	✓	✓	✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6	✓	✓		✓

References:

1. Introduction to Mathematical Statistics, Hogg R V Craig A T, Macmillan
2. Mathematical Statistics, Freund J E, Waple R E, Prentice Hall of India.
3. Probability and Statistics for Engineers, Miller I Freund J E, Prentice Hall of India.
4. Statistics for Management, Levin R I, Prentice Hall of India
5. Introduction to Mathematical Statistics, Hogg R V Craig A T, Macmillan
6. Mathematical Statistics, Freund J E, Waple R E, Prentice Hall of India.